Solution Key: The Rotating Crane Arm

Chapter 10: Rotational Dynamics | Grade 11 Scientific

Part 1: Static Equilibrium

1. Force Analysis

The three external forces acting on the rod OB are:

1. **Weight ($\vec{W}$):** Applied at the Center of Gravity (G), which is the midpoint of the rod ($1\,m$ from O).
2. **Tension ($\vec{T}$):** Applied at the extremity B ($2\,m$ from O).
3. **Reaction ($\vec{R}$):** Applied at the pivot axis passing through O.

2. Moments & Equilibrium

Condition of Rotational Equilibrium: The sum of the moments of all external forces with respect to the axis of rotation must be zero:

$$\sum M_\Delta(\vec{F}_{ext}) = M_\Delta(\vec{W}) + M_\Delta(\vec{T}) + M_\Delta(\vec{R}) = 0$$

Moment of Weight ($M_\Delta(\vec{W})$): The weight acts at the midpoint ($L/2 = 1\,m$). In the horizontal position, the force is perpendicular to the lever arm. Since it tends to rotate the rod Clockwise (negative direction):

$$M_\Delta(\vec{W}) = -(W \times L/2) = -(Mg \times L/2) = -(3 \times 10 \times 1) = \mathbf{-30\,N \cdot m}$$

3. Calculating Tension

The reaction $\vec{R}$ passes through the pivot, so $M_\Delta(\vec{R}) = 0$.
The tension $\vec{T}$ acts at B ($L = 2\,m$) and tends to rotate the rod CCW (positive):

$$M_\Delta(\vec{T}) = +(T \times L) = 2T$$

Applying equilibrium:

$$-30 + 2T = 0 \implies 2T = 30 \implies T = \mathbf{15\,N}$$

Part 2: Rotational Dynamics

4. Conceptual Understanding

Angular Speed (ω): Both points have the **same angular speed** because the rod is a rigid body; all points sweep through the same angle in the same time interval.
Linear Speed (v): They have **different linear speeds**. Since $v = r\omega$, the speed depends on the distance from the pivot. Point O has $v = 0$ (at pivot), while point B has the maximum speed $v = L\omega$.

5. Moment of Inertia

$I_{total} = I_{rod} + I_{block}$

- $I_{rod} = \frac{1}{3}ML^2 = \frac{1}{3}(3)(2^2) = 4\,kg \cdot m^2$
- $I_{block} = m(L)^2 = (1)(2^2) = 4\,kg \cdot m^2$

Total Inertia: $4 + 4 =$ **8 kg · m²**

6. **Net Torque Calculation**

At the instant the cable is cut, the rod is horizontal. Both the rod's weight and the block's weight act downwards (CW rotation):

- $M_{W,rod} = -30\,N \cdot m$ (Calculated in Part 1)
- $M_{W,block} = -(m \cdot g \times L) = -(1 \times 10 \times 2) = -20\,N \cdot m$

$$\sum M = -30 - 20 = -\mathbf{50\,N \cdot m}$$

The magnitude is $50\,N \cdot m$ in the clockwise direction.